Static Discharging – General Data

1. General

Static discharge wicks are installed as follows: One on each aileron, one on each outboard fore flap hinge arm assembly, two on the right hand elevator, one on the left hand elevator and three on the rudder. The wicks serve to dissipate electrostatic charges from the aircraft structure and thereby minimize radio interference from this source. The static discharge wicks consist of a rope floss of fine fibers about 13 inches long. For mechanical protection the wick is encased in a plastic sheath about one-quarter inch in diameter. One end is fixed to an electrical lug which in turn is secured to the airfoil by two screws and lock washers. At the other end of the plastic sheath, the rope floss extends about one inch so that the fibers are free to ionize in the slipstream by virtue of the impressed potential between the electrified aircraft and the surrounding space.

2. Servicing

A. Service Wicks

1) Cut off ragged edges from wick. Clean if necessary with lint-free cloth.

2) Measure one inch from exposed end of rope floss, and mark plastic sheath.

3) Carefully cut back and remove plastic sheathing to position marked in Para 2.A.2), and remove the cotton covering from around the rope floss.

4) Ensure that the overall length of serviced wick exceeds six inches.

3. Removal/Installation

A. Remove Static Discharge Wick

1) Remove two screws and lock washers securing static discharge wick to airframe, and remove the wick.

B. Install Static Discharge Wick

1) If necessary, prepare the end of the new static discharge wick, using procedure described in Para 2.A.

2) Secure new wick to airframe with screws and lock washers, making sure that metal to metal contact is maintained. Refer to Chapter 20-75-00 for electrical bonding requirements.

3) Ensure parallel alignment with line of flight.

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